

Problem 3

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Problem 1 Find the x -coordinates of global extrema and global extreme values of f on the given interval.

(a) $f(x) = x\sqrt{2-x^2}$ on $[-\sqrt{2}, \sqrt{2}]$.

(b) $f(x) = x^3e^{-x}$ on $[-1, 5]$.

(c) $f(x) = x \ln\left(\frac{x}{5}\right)$ on $[0.1, 5]$.